

Demographic dynamics in fluoride contaminated regions: a comparative analysis with Islamabad, Pakistan



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CIIT/SP21-BSO-062/ISB

BS Thesis

In

Biosciences

COMSATS University Islamabad, Pakistan

Fall, 2024



COMSATS University Islamabad

**Demographic dynamics in fluoride contaminated
regions: a comparative analysis with Islamabad,
Pakistan.**

A Thesis Presented to

COMSATS University Islamabad

In partial fulfilment

of the requirement for the degree of

BS (Biosciences)

By

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Demographic dynamics in fluoride contaminated regions: a comparative analysis with Islamabad, Pakistan.

An Undergraduate Thesis submitted to the Department of Bioscience as partial fulfillment of the requirement for the award of Degree of BS (Biosciences).

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Dedicated To My Beloved Parents

Whose prayers, affection and encouragement made me able to get success.

ACKNOWLEDGMENTS

This thesis is dedicated to my beloved family, whose unwavering love, support, and encouragement have been the cornerstone of my success. Their belief in me has been a source of strength throughout this journey, and I am eternally grateful for their constant presence in my life.

ABSTRACT

This study assesses the occurrence of dental fluorosis in the Punjab province of Pakistan and focuses especially on the data of Lahore and Mianwali districts with Islamabad as a control. Groundwater contamination by fluoride in these regions often exceeds levels recommended by the World Health Organization and poses a significant public health problem. Stratified random sampling was used to give a representation of demographic diversity on the group surveyed, and a cross sectional survey was conducted among schoolchildren aged 12–16 years. Dietary habits were assessed by structured questionnaires, and fluorosis severity was assessed by a structured dental examination to evaluate Dean's Index. Results revealed significant disparities in the prevalence of dental fluorosis, with Islamabad exhibiting the lowest rates of fluorosis across all categories (Normal: 57 cases), compared to Mianwali (Normal: 30) and Lahore (Normal: 11). Mianwali and Lahore presented higher instances of questionable, very mild, mild, and moderate fluorosis, with Lahore having the highest number of moderate fluorosis cases (21). Severe fluorosis was observed primarily in Lahore (6 cases) and minimally in Mianwali (2 cases). Statistical analysis showed a significant variation (P value = 0.001) in fluorosis distribution across the regions. This study highlights the urgent need for targeted interventions and sustainable water management practices to address fluoride contamination and mitigate the health risks posed to vulnerable populations in fluoride-endemic areas.

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CHAPTER 1:

INTRODUCTION

1.1. Background:

Fluoride is an incredibly special chemical that is both essential in trace amount and toxic when present in certain concentration and hence has attracted a lot of attention from researchers and public health experts. Fluoride in measured, controlled dosages act as a promoter of growth thereby enhancing teeth's hardness, avoidance of decay, and general body skeletal frame improvement. In the increased environmental fluoride concentration zones, it is a potential hazard in rural/peri-urban areas around Islamabad contributing to chronic fluorosis while it is an important nutritional micronutrient. Comparative studies indicate that fluoride exposure is highly variable across the country and Islamabad as a developing urban center is a good case to study these demographic dynamics. (Tahir & Rasheed, 2013)

Fluoride is introduced into the population in the form of fluoride-contaminated drinking water, dental products, salt, and other community level preventive measures in fluoride-contaminated regions including, for example, around Islamabad. Usually recommended is a range of fluoride in drinking water concentration from 0.7 up to 1.5 mg/L, depending on climatic conditions and drinking water consumption (Rahman, Khan, Ahmad, Mian, Saeed, Afaq, Khan, Smith, Mian, April-June 2018). Geological factors, however, make it the case that much groundwater in rural and peri urban areas surrounding Islamabad have fluoride concentrations more than safe levels, and hence the exposure of residents there is chronic. (Ling et al., 2022)

Fluoride exposure does not descend evenly unto the population in Islamabad. Fluoride toxicity studies have shown that children are more prone to toxic levels of fluoride than adults and more so during their physical, or cognitive development age. However, analysis has shown that chronic exposure in these formative years is linked with dental and skeletal fluorosis and reduced cognitive function — including IQ levels. As an example, in fluoride endemic regions children are at a greater risk of central nervous system (CNS) dysfunctions according to Green et al. (2020). But it's even more alarming since Islamabad is reeling under an increasing number of children and adolescents in fluoride affected zones (Sohail et al., 2020).

Also important are gender and socioeconomic status. Preponderances of this risk exist in women especially during lactation and pregnancy, with increased calcium and fluoride metabolism. (Dewey et al., 2023)

Women in Islamabad's peri-urban areas, where there is limited access to adequate healthcare and nutrition, are disproportionately worse affected. Just as dietary patterns, including reliance on groundwater rich in naturally occurring fluoride for cooking and drinking, accelerate their susceptible populations' exposure levels.

In contrast to other fluoride-contaminated regions for which community based interventions — such as defluoridation of drinking water and targeted nutritional programs — have been implemented to mitigate health risks, Islamabad's situation is unique. In Islamabad, some interventions exist but they have a narrow reach and don't help. Early detection followed by an awareness campaign and alternate sources of water is necessary to prevent fluorosis and other fluoride related health problems in the region. (Nikhil Mankar et al., 2023)

Finally, the demographic complexity of fluorides is demonstrated in the case of Islamabad. Just as urgent is the need for localized interventions in response to the region's combination of urbanization, dietary patterns and groundwater contamination. To fully benefit from fluoride while minimizing its adverse effects, in particular, among children, women, and poor populations are, an approach of balance is required. These demographic dynamics can be used as a model for any fluoride contaminated regions globally.

1.2 Introduction to Fluoride:

Fluoride is an ion of fluorine which is a very useful micronutrient for humans but a dangerous pollutant for environment when present in large concentrations. As a highly soluble and mobile ion, F^- released through natural geological and human activities has potentially disastrous impacts on both the environment and human health. At present about 211 million people in over 35 countries are affected by fluorosis which is a disease arising from excessive take of fluoride. This integrated global problem underlines the necessity for further studies, applications of ecological solutions, and focused prevention and treatment strategies of the fluoride contamination and its impacts. The effect of intake of fluorides is not the same, but depends on a range and variety of factors such as age, sex, social-economic status, genes and the diet level of the community. For instance, dental fluorosis arises in children because their teeth are still developing, while the risk of skeletal fluorosis is considerably higher in those who have a poor diet. All these factor contribute to the fluoride sensitivity and the severity of the adverse health effects.

By understanding them within the domain of lag-time and threshold, the two implementations of fluoride make sense. It is against this backdrop of escalating fluoridation, increased use of a

toxic pollutant and the emergence of an evolving developing urban landscape that Islamabad provides a fascinating study for investigating these dynamics. Such comparisons prove that there is high fluctuation in the composite level across the globe and by therefore, the differences which exist in the geographical and social demography in defining the conditions related to fluorides. To solve for this necessity, it is important not only to focus on the problem of fluoridising water and regulating the presence of fluoride in it but also on the issue of vulnerability with all their socio-economic and environmental determinants.

1.3. Naturally Occurring and Artificial Fluoride:

Fluoride enters drinking water through two main pathways:

- **Naturally Occurring Fluoride:** Fluoride occurs naturally in water well water being the most affected by geographic formations, hence its concentration is adjusted to the region's soil formation.
- **Artificial Water Fluoridation:** This is achievable by the fluoridation of water that is public water supply as a measures of preventing tooth decay.

Whereas, EPA deals with the naturally occurring fluoride in the drinking water by setting MI value for fluoride at 4.0 mg/L and the recommended AL value is 2.0 mg/L in order to protect the health of the citizens, the optimal concentration of Fluoride for dental health is set by the US Public Health Service (PHS). In 2015 the PHS advised maintaining a fluoride concentration of 0.7 mg/L in the community water systems, as it has been deemed both effective in cavity prevention, and is not likely to cause dental fluorosis.

They do this without overlooking some issues such as climate, water usage, and other sources of aldosterone system with fluorides such as paste. Whereas, the EPA's rules address nutritional fluoride and seek to avoid cases of overexposure, the PHS guidelines are intended to strike the right proportion of exposure and all the benefits that go with it.

1.3. Benefits and Risks of Fluoride:

Since it is consumed in small quantities, Fluoride has useful effects on dental health due to its ability to make teeth harder and resistant to decay. But overuse of this chemical may lead to very adverse effects on human health. The main root of fluoride intake by the human food

chain is through the water, with most ground water supplies being a source of major concern particularly in the third world where water is mostly taken raw.

The WHO has prescribed 1.5 mg/L as the safe level of fluoride in drinking water. Tooth and skeletal fluorosis occur when a person's intake rises beyond this level, with symptoms that include mottling of the teeth, calcification of ligament and progressive deformities in the bones. High fluoride concentrations have also been associated with cancers, damage to the nervous system and impaired brain function in children and mentally retarded individuals (Kowalski, 1999; Li et al., 1994; Yiamouyiannis, 1993).

1.4. Fluoride in Tropical and Semi-Arid Regions:

The issue of fluoride contamination is particularly severe in tropical and semi-arid regions. In these areas, intense weathering of rocks under warm climatic conditions naturally increases fluoride concentrations in groundwater. Moreover, high water consumption which is common in humid regions also enhances the toxicity of fluoride and increases the population exposure to the deal health effects (Dissanayake, 2005; WHO, 1996, 2004).

As such, there is a pressing need to review the Aql limits of fluoride in drinking water to such regions which have high water intake. The appropriate complex water treatment systems, integration with the key public health measures are the primary solutions to minimize the causes of fluoride and its consequences on vulnerable populations.

1.5. Geochemistry and Hydrochemistry of Fluoride:

In this study sources, nature, concentration and mobility of fluoride in groundwater has been determined with the help of geochemical and hydrochemical parameters such as pH, water temperature and water to rock interaction (Kitalika et al., 2018). Wang et al. (2013) and Warren et al. (2005) reported that there positive relationship between fluoride and the other ions such as chloride, sodium, bicarbonate, and boron, TDS, pH, and E.C.; however, there is an inverse relationship between the amount of fluoride and calcium ions (Ca^{2+}) (Karro and Uppin, 2013). The provided correlations provide information on dissolution processes and sources of fluoride. For example, in Northern China, Wang et al. (2013) reported that F had positive correlations with Cl and HCO_3 though there was similar observation in northern Argentina by Warren et al. (2005). In the carbonate aquifer system of Estonia the above two ions also showed positive nature of association may be because both fluoride and boron dissolve in the water from the

carbonate rocks (Karro and Uppin, 2013). In the Dakota aquifer system of the USA by Karro and Uppin (2013), fluoride had a direct relationship with bicarbonate, pH, TDS and electrical conductivity, but calcium had an inverse relationship with fluoride. It enables one to evaluate the fluorspar contamination sources for assessing fluoride solubility in ground water.

1.6. Fluoride Contamination in Pakistan and Global Implications:

Fluoride contamination, particularly at levels exceeding the World Health Organization's suggested maximum of 1.5 mg/L, is common in certain areas of Pakistan where concentrations are even rising. They are mostly rural areas with a variety of challenges, poor access to healthcare, substandard water treatment facilities, and little public exposure to the hazards of fluoride poisoning.

Excessive intake of fluoride, a direct consequence of which is fluorosis, is a global health problem that afflicts millions of people in Asia, Africa, South America and Australia. There are over 62 million people who consume water with excess fluoride in India. The problem is compounded by rapid urbanization, industrialization and poor water handling/management practices.

1.7. Water Quality in Pakistan:

1.7.1. Overview of water quality issues in Pakistan:

Polluted water in Pakistan is the product of both natural circumstances (high mineral geological structures) and human actions (agricultural runoff, industrial discharge and poor sanitation). The country is buffeted with contamination threats from both biological and chemical pollutants.

Fluoride is one of the major pollutants that affect water quality in Pakistan. The fluoride levels in groundwater exceed safe limits suggested by the World Health Organization (WHO) in parts of Pakistan that have lead to a serious threat to health. Both natural (from leaching of fluoride bearing minerals) and anthropogenic (industrial waste discharge and agricultural practices) contamination contribute to the fluoride loading. These findings indicate the need for robust help mitigating fluoride in groundwater and it's damaging influence on human health and ecosystems.

1.7.2. Link to fluoride contamination challenges:

Pakistan, especially its rural areas, have a critical water quality problem – fluoride contamination. Groundwater locations where fluoride naturally occurs in the water include areas where fluoride is added to drinking water, such as in parts of Western Europe; and groundwater locations where fluoride is further worsened by human activity such as use of excess fertilizers and industrial practices.

In Pakistan, fluoride concentration in drinking water may exceed 1.5 mg/L, the level exceeded by WHO and defined as unsafe for consumption in water. Dental and skeletal fluorosis resulting from extended exposure to high fluoride levels is a serious public health issue.

1.8. Water Quality Challenges in the Mianwali Region:

1.8.1. Mianwali's geography and geology:

Mianwali city is situated in Punjab province of Pakistan where availability of water is a serious problem; in this regard, this semi-arid region has no external surface water source but required essential groundwater source.

1.8.2. Current fluoride levels and mapping hotspots:

Groundwater of Mianwali has high fluoride concentrations and many areas have been found to exceed WHO's safe fluoride limit. The level of fluoride contamination varies from place to place; however, it becomes more pronounced in the area of the region in the southern and eastern parts which are labeled as hotspots. Addressing the issue requires mapping these hotspots.

1.8.3. Sources of contamination:

Natural fluoride contamination is the main source of fluoride contamination in Mianwali district as formed by the geological formations in the area. In selected areas, fluoride can be absorbed into water from fluoride containing minerals. The major reason for contamination, however, is from natural sources, and although natural sources are the major reason for contamination, improper water management practices and over extraction of groundwater can contribute to it.

1.9. Fluoride Exposure in the Mianwali Region:

1.9.1 Groundwater dependency in the Mianwali region:

Mianwali, like much of rural Punjab, relies heavily on groundwater for drinking water. The region has few surface water resources, and the availability of groundwater makes it a vital source. However, due to high fluoride concentrations in many areas, this dependence poses health risks to the population.

1.9.2. Data analysis on fluoride concentration in water samples:

Studies analysing water samples from Mianwali show significant variation in fluoride levels across different areas. Some regions have concentrations well above the safe limit, with concentrations ranging from 2.5 to over 5 mg/L. Data analysis has indicated that areas with deeper wells or limited filtration often have higher levels of fluoride.

1.9.3. Impact on rural vs. urban populations:

In Mianwali, rural populations are disproportionately affected by fluoride exposure. Most rural areas rely on local wells, which are often contaminated with high levels of fluoride. In contrast, urban areas may have more access to water treatment facilities, although some urban residents still depend on untreated groundwater sources. As a result, the rural population is more vulnerable to the health impacts of fluoride exposure, such as dental and skeletal fluorosis.

1.10. DENTAL CARIES:

Dental caries is a major disease that occurs due to the multifactorial causes leading to the hardening of tissues of the teeth. It majorly causes tooth loss and has been seen amongst various regions around the world.

Fluoride is crucial for controlling dental plaque and maintaining a balanced diet. A large number of studies have identified various molecular mechanisms by which pathogenic microorganisms have the ability to increase the biomass of oral biofilm in the presence of sucrose, and the ability to promote biological changes of the oral environment, causing dental caries.

In addition to these factors, there are other external factors. Socio-economic and behavioural factors may influence the development of dental caries. The age, gender, scholarship, the presence of fluorides, oral hygiene habits, among others are all factors that influence the development of dental caries. (Nélio Veiga¹, 2016)

1.10.1. Impacts on health:

Despite a decline, recent studies indicate a significant increase in almost all age groups of people affecting both primary and permanent teeth. But most of all the caries are highlighted in the region that are not living up to the standards ie mostly belonging to the lower socioeconomic groups, children or even new immigrants. Factors contributing to the rise in caries may include the use of bottled water over fluoridated tap water and dietary changes. The influx of immigrants and the movement of rural populations to urban centers in search of jobs have led to negative changes in diet, lifestyle, and health, contributing to increased caries. ROBERT A.BAGRAMIAN, DDS, MPH, PHD, FRANKLIN GARCIA-GODOY, DDS, MS & ANTHON R.VOLPE, DDS, MS)

1.10.2. Longitudinal Trends of Fluoride Concentration:

Fluoride is widely recognized for its role in dental health; however, its excessive intake can pose significant health risks, particularly in children. Despite its ubiquity in fluoridated salt, natural water, and dietary sources, the long-term metabolic consequences of fluoride exposure during early developmental stages remain poorly understood. Some studies suggest that early-life fluoride exposure disrupts cardiometabolic health, with potential long-term consequences. Mechanisms may include oxidative stress, hormone dysregulation, and lipid metabolism disturbances. Sex-specific effects further underscore the need to examine vulnerable subpopulations. These results challenge the assumption of fluoride's safety as a universal public health intervention and call for critical re-evaluation of community fluoridation programs, especially in regions with high salt and fluoride consumption like Mexico.(Aldana et al., 2023)

1.11. The Effect of Socioeconomic Status on General Health:

Socioeconomic status is the social and economic level of an individual or family in his/her society. This status has a huge influence on health, education and life experiences (Nutakor et al., 2023).

Below are the major contributing factors to socioeconomic status and their general health impacts

- Income

Income is an essential determinant of a person or families social class status. Often, income can provide an indication of the economic status provided by the population, imputes where necessary given the yearly spending. Health being a component of wealth, the poor are know to suffer higher levels of diseases and premature mortality (Marmot et al., 2008).

Burbach discussed that low income comes with some social drawbacks like; Early drop out on education, insecure housing, poor or unsafe living conditions, and unfavorable employment. They produce psychosocial and environmental demands that influence health, with higher likelihoods of boosted morbidity and mortality (Boylan et al., 2018). The above adverse conditions cause stress leading to physiological changes of catecholamines, cortisol, and blood pressure that increase cardiovascular disease risk (Cuevas et al., 2018).

- Education

Education is closely linked to both life span and sickness rates (Marmot et al., 2008). For instance, the life expectancy of a college learner is believed to be ten years longer than that for a high school dropout (Case & Deaton, 2021). The level of education is also a critical predictor of Cardiovascular Diseases (CVD). Meta-analysis of data show that education decreases hypertension risk and subsequently CVD (Dégano et al., 2017).

Education determines and is determined by the social economy including employment, income, and appropriation. Education is inversely related to living and working conditions; less education leads to exposure to such stressors as unemployment, violence among others; and less resources to handle stress. Such factors make people develop these complications most of which are life threatening like hypertension (Williams et al., 2016).

- Employment Status

Employment status has been found to have a major effect on physical and psychological well being (Marmot et al., 2008). Contact with toxic agents at work and job insecurity compromise the status of personal health. The Health Department (2012) states that people experience loss of health insurance, and increased psychological stress when they lose their job.

This means that, when a person loses their job they can no longer afford to pay for coverage and do not qualify for government funding either. This barrier therefore leads to poor health and decreased access to health care services (Silver et al, 2021). Some employment aspects such as status continue to reflect SES and determine health state proximal to work and employment (Schultz et al., 2018).

- Aims and Objectives

Therefore, the objective of this study is to establish the prevalence of dental fluorosis in a targeted population and assess the dietary profile and correlation affecting dental fluorosis.

- Determine the concentration of fluoride in Lahore's Punjab ground water.
- Estimate the degree of fluorosis in the children of the selected study area by performing dental examinations according to the set procedures.
- It allows examining the patterns of association between socioeconomic characteristics and the level of fluoride exposure in the target population.
- Compare and contrast the pattern of dietary factors with dental fluorosis.

Chapter 2:

Material and Methods

2.1. Study Area

The research was conducted in Lahore and Mianwali district of Punjab province, Pakistan. These districts were specifically chosen due to their classification as endemic regions for dental fluorosis (*Ling et al., 2022*). The rural landscape and agricultural economy of these areas play a significant role in shaping the dietary habits of the local population. A majority of the residents depend on groundwater as their primary drinking water source, which has been reported to contain fluoride levels exceeding the World Health Organization's recommended limits. This makes the region a hotspot for dental fluorosis. The study focused on several schools across the districts to ensure diverse representation of the younger population. Schools were selected strategically based on their geographical location and accessibility, allowing for efficient data collection. Conducting surveys in schools provided a controlled environment to analyze children, whose developing teeth and bones are particularly vulnerable to fluoride exposure, and whose dietary habits could be effectively monitored.

2.2. Data Collection

Data collection was carried out in Lahore, Mianwali District Punjab, Pakistan and Islamabad was chosen as a control group. The study targeted schoolchildren aged 12 to 16 years from grades 7 to 10. A stratified random sampling method was employed to select participants, ensuring representation across different demographic and socioeconomic groups within the village. Stratification criteria included age, grade level, and gender to achieve a balanced and comprehensive sample for analysis.

2.3. Questionnaire Design

A structured and detailed questionnaire was designed to evaluate participants' dietary habits and related factors contributing to dental fluorosis.

Section One: Demographic Information

- **Age:** To categorize participants into specific age groups and analyze age-related patterns in dental fluorosis prevalence.

- **Height and Weight:** Measurements were taken to assess participants' nutritional status, which was later used to calculate body mass index (BMI) and explore its correlation with dental health.
- **Family Size:** Participants were asked about the number of family members to examine family size as a variable influencing dietary habits and overall household health dynamics.

Section Two: Dietary Habits

- **Milk Consumption:** Questions were designed to assess the frequency and quantity of milk intake, as calcium is known to influence dental health.
- **Egg Consumption:** Participants provided details about their egg consumption frequency, as eggs can contain fluoride depending on the chickens' diet.
- **Tea Consumption:** Given that tea leaves tend to accumulate fluoride from the soil, participants were asked about their tea consumption habits, including frequency and serving sizes.

2.4. Examination of Dental Fluorosis

A qualified dentist conducted dental examinations to assess the severity of dental fluorosis among participants. The severity was classified using Dean's Index, which categorizes fluorosis into six levels: Normal, Questionable, Very Mild, Mild, Moderate, and Severe.



2.5. Study Duration

The study was conducted over a five-month period, from August 2024 to December 2024, in Lahore, Mianwali District keeping Islamabad as a control group. This carefully selected time frame allowed for thorough data collection and analysis, accounting for agricultural cycles and seasonal variations that could influence dietary habits and, consequently, the prevalence and severity of dental fluorosis.

2.6. Ethical Approval

Ethical approval for the research was obtained from the Ethics Review Board of the Department of Life Sciences, COMSATS University Islamabad (CUI). The study strictly adhered to established ethical guidelines to ensure the well-being, confidentiality, and informed consent of all participants, thereby upholding the integrity of the research process.

CHAPTER 3

RESULTS

1.1. PREVALENCE OF DENTAL FLUOROSIS IN PAKISTAN:

The cross-tabulation table analysing the prevalence of dental fluorosis, across three locations:

Lahore, Mianwali, and Islamabad, reveals significant insights. Among the observed categories, the number of individuals with normal dental fluorosis is highest in Islamabad (57), followed by Mianwali (30) and Lahore, totalling 11 cases.

Questionable fluorosis exhibits a more dispersed pattern, with the highest count in Mianwali (12), followed by Lahore (7), and the least in Islamabad(3) Very mild fluorosis present in Lahore(9) and Mianwali(11), while Islamabad reports (1). Mild fluorosis is more common, especially in Mianwali (4) and Lahore (3) , with a lower count in Islamabad (1). The most pronounced moderate fluorosis cases are in Lahore (21), followed by Mianwali (4), with no cases reported in Islamabad, making it the most prevalent category across all locations

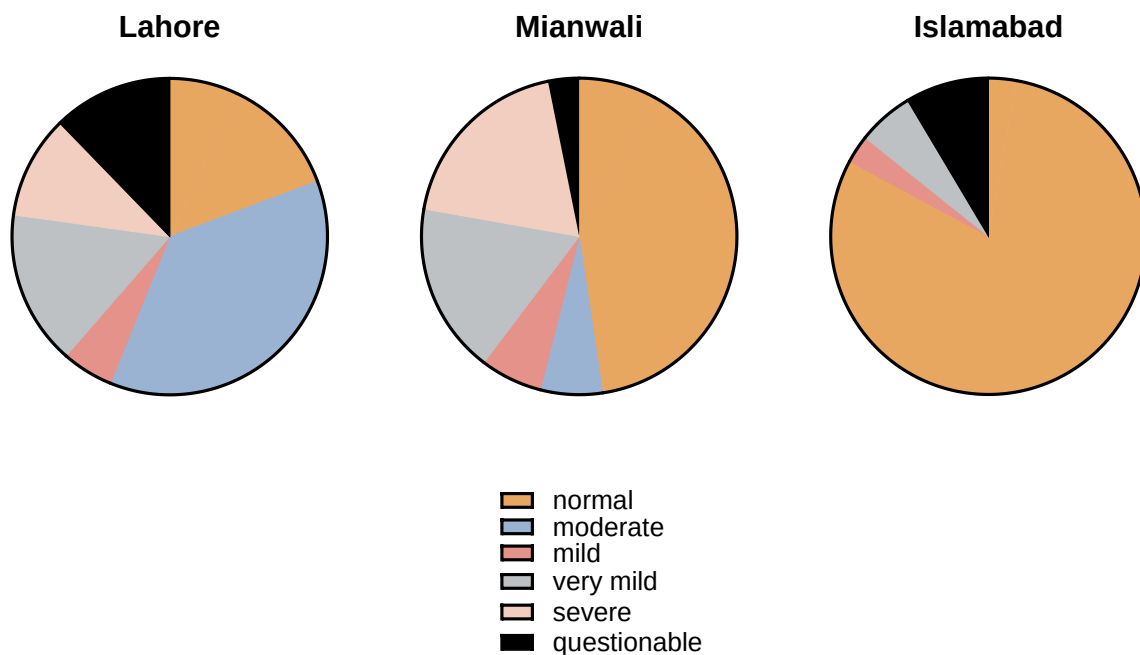
Severe fluorosis is observed significantly in Lahore (6) and minimally in Mianwali (2), with none in Islamabad, resulting in 17 cases overall. The total number of cases across all fluorosis categories sums to 184, with Lahore presenting the highest overall incidences (46), followed by Mianwali (29) and Islamabad (6).

The statistical analysis yields a P value of 0.001, indicating a significant variation in the distribution of dental fluorosis among these locations.

1.2. Dental Fluorosis Across Cities:

This table compares dental fluorosis levels in three cities—Islamabad, Lahore, and Mianwali. Islamabad has the highest number of individuals in the "Normal" category, with 57 cases, while Lahore stands out for having more individuals with "Moderate" fluorosis (20) and a smaller number in the "Very Mild" category (9). Mianwali shows a more evenly distributed pattern of fluorosis severity, with a notable count in the "Questionable" category (11).

Crosstab					
		City			Total
		Islamabad	Lahore	Mianwali	
Dental Fluorosis	Normal	57	11	31	99
	Moderate	1	20	4	25
	Mild	1	3	4	8
	Very Mild	2	9	11	22
	Severe	0	6	2	8
	Questionable	3	8	11	22
Total		64	57	63	184



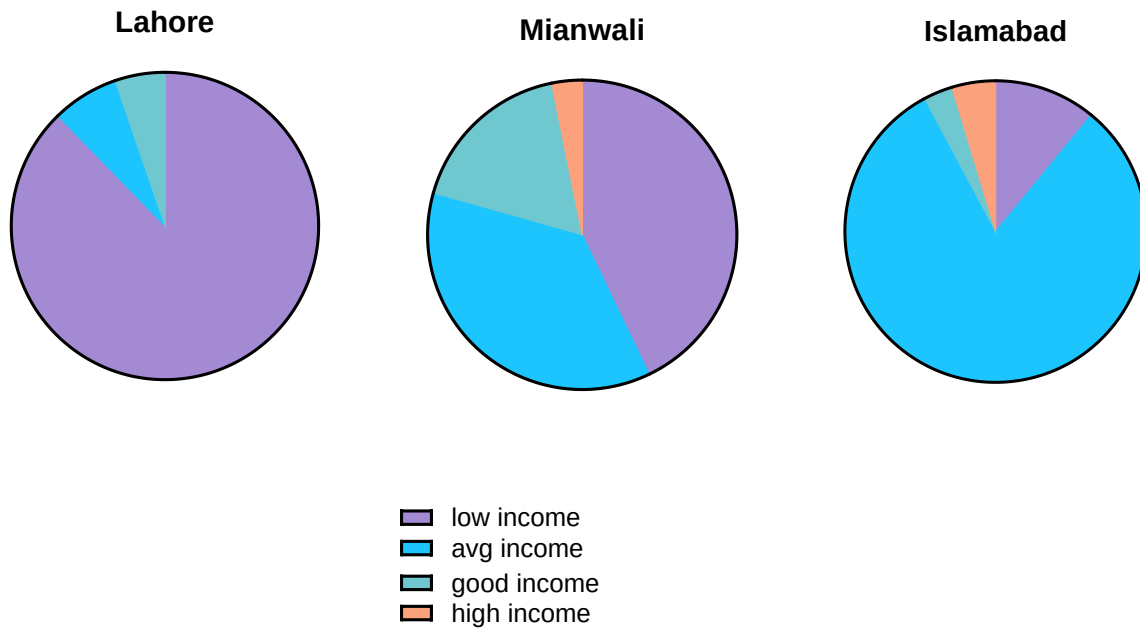
1.3. SOCIOECONOMIC FACTORS ASSOCIATED WITH FLUORIDE EXPOSURE:

3.3.1. FAMILY INCOME:

This table shows family income patterns. Islamabad has a significant number of families in the "Average Income" group (52), while Lahore has more families in the "Low Income" group (49). Mianwali has a more balanced spread between "Average" (23) and "Good Income" (10). As the P value is less than 0.05 there is a significant association between income and fluoride exposure

TABLE:

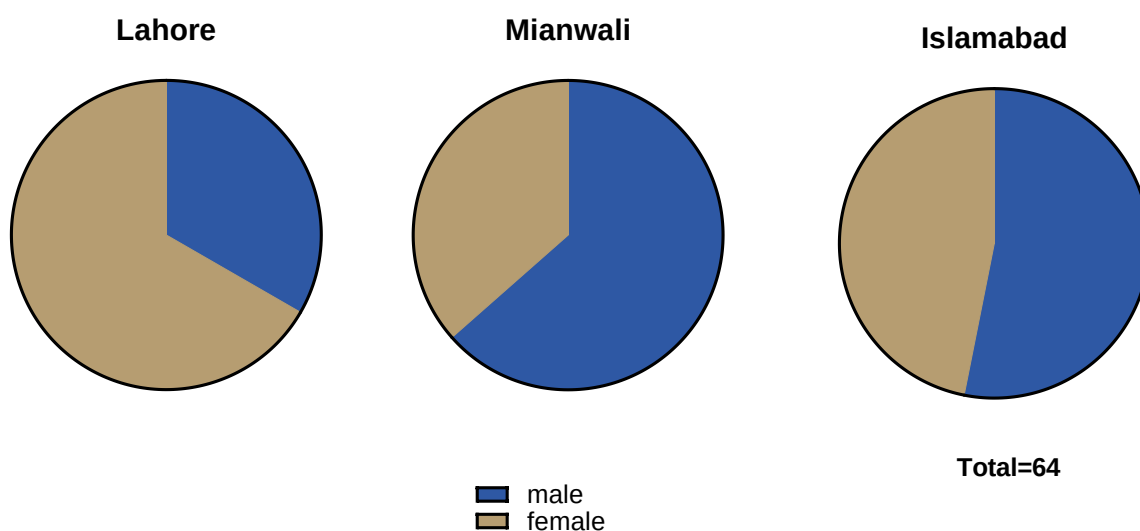
City * Income Crosstabulation						
		Income				Total
		Low Income (1-30k)	Average Income (30-70k)	Good Income (70-100k)	High Income (100-250k)	
City	Islamabad	7	52	2	3	64
	Lahore	49	4	4	0	57
	Mianwali	28	23	10	2	63
Total		84	79	16	5	184
P Value						0.001



3.3.2.GENDER:

This breakdown by gender reveals interesting trends. In Islamabad, males and females in the "Normal" category are roughly equal. However, Lahore shows a higher number of males experiencing "Moderate" fluorosis. In Mianwali, both genders show a strong presence in the "Questionable" category, with males having a slight dominance in the counts.

Level of Dental Fluorosis * Gender Crosstabulation								
		Gender						Total
		Islamabad		Lahore		Mianwali		
		Male	Female	Male	Female	Male	Female	
Level of Dental Fluorosis	Normal	28	18	3	13	16	15	93
	Moderate	0	0	12	9	2	2	25
	Mild	1	0	1	2	3	1	8
	Very Mild	2	0	0	10	8	3	23
	Severe	0	0	2	4	2	0	8
	Questionable	3	4	1	8	8	3	27
Total		34	22	19	46	39	24	184

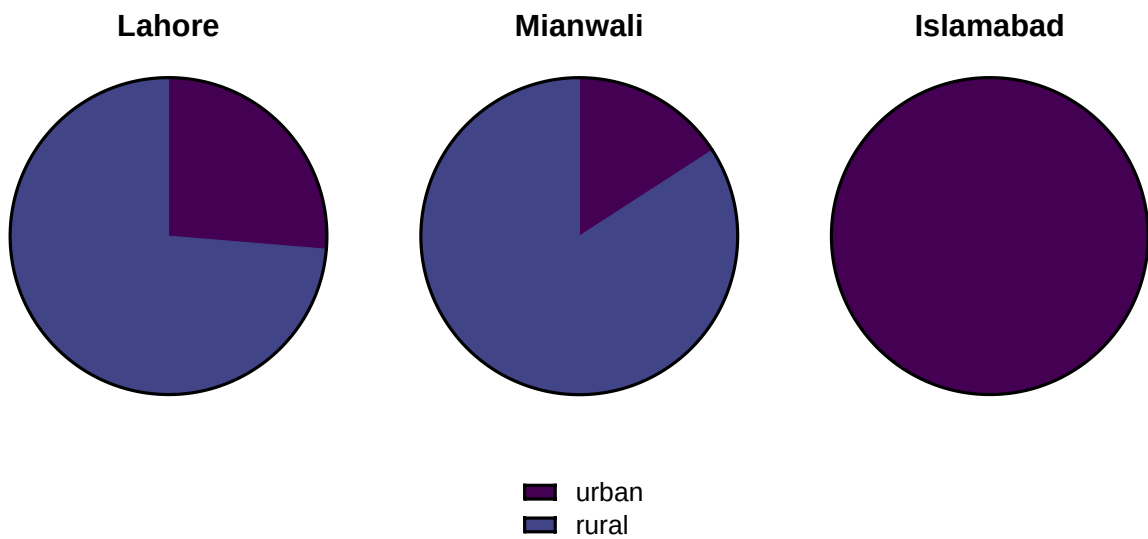


3.3.3. AREA OF RESIDENCE:

Islamabad, being entirely urban, has a large number of people with "Normal" fluorosis. Lahore's rural areas show more cases of "Moderate" fluorosis, with a noticeable rural presence in Mianwali's "Questionable" category. The statistical analysis yields a P value of 0.991, indicating there is no significant association between area and fluoride exposure

Level of Fluorosis * Area Crosstabulation								
		Area						Total
		Islamabad		Lahore		Mianwali		
		Urban	Rural	Urban	Rural	Urban	Rural	
Level of Fluorosis	Normal	63	0	10	4	1	25	104
	Moderate	1	0	2	19	1	3	26
	Mild	2	0	1	2	0	4	9
	Very Mild	3	0	1	8	3	8	24

	Severe	1	0	2	4	0	2	9
	Questionable	4	0	4	4	3	8	28
Total		74	3	20	41	8	50	184
P Value								0.991

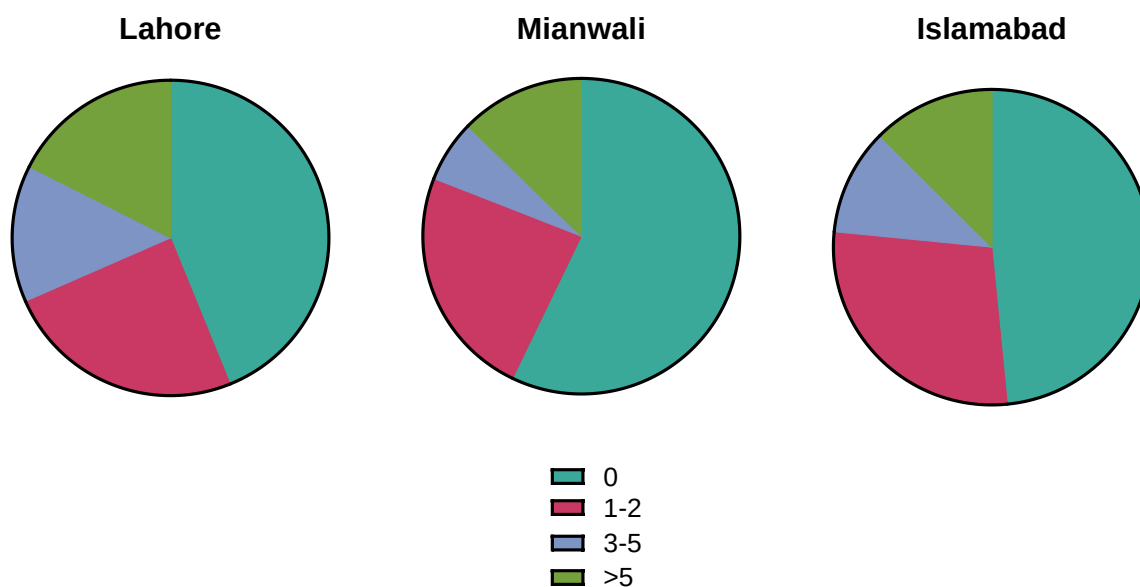


3.4. DIETARY HABITS ASSOCIATED WITH FLUORIDE EXPOSURE

3.4.1. Milk intake:

Milk intake patterns vary across cities. Islamabad has the highest number of individuals who consume no milk (31), while Lahore and Mianwali show a more balanced spread across moderate and higher intake levels. As the P value is more than 0.05 there is no significant association between milk intake and fluoride exposure.

city * Milk Intake Cross tabulation						
		Milk Intake				Total
		0	1-2	3-5	>5	
city	Islamabad	31	18	7	8	64
	Lahore	24	15	8	10	57
	Mianwali	37	15	3	8	63
Total		92	47	18	26	184
P Value						0.492

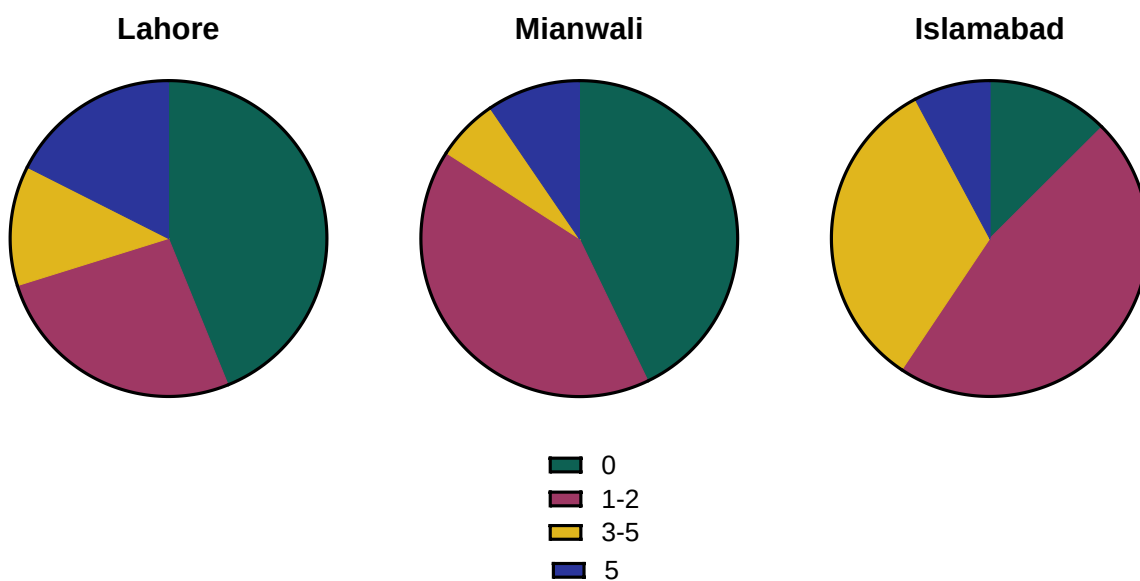


3.4.2. YOGURT INTAKE

Yogurt consumption is highest in Islamabad, where most individuals consume 1-2 servings a week (30). Mianwali, on the other hand, has a significant number of people not consuming yogurt at all (27), while Lahore shows a balanced distribution. As the P value is less than 0.05

there is a significant association between yogurt intake and fluoride exposure of the different cities of Pakistan.

city * yogurt Intake Crosstabulation						
		Yogurt Intake				Total
		0	1-2	3-5	>5	
city	Islamabad	8	30	21	5	64
	Lahore	24	15	8	10	57
	Mianwali	27	26	4	6	63
Total		59	71	32	21	184
P Value						<.001

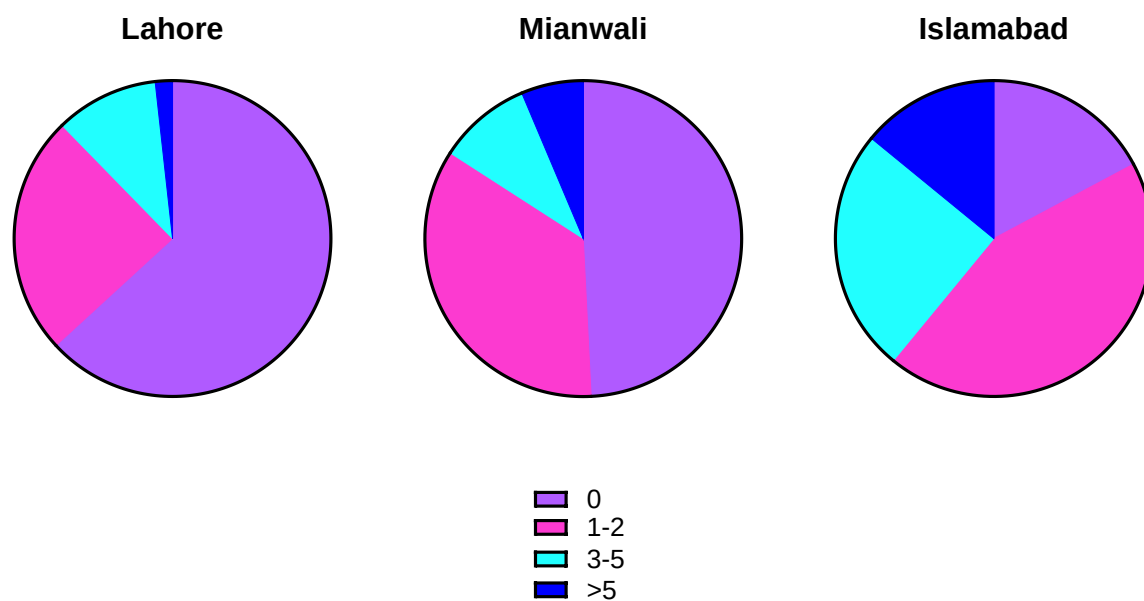


3.4.3. NUMBER OF EGGS CONSUMED:

Egg consumption is most common in Islamabad, where 28 people consume 1-2 eggs. In Lahore, 35 people report not consuming eggs, while Mianwali's largest group consumes 1-2

eggs (22). As the P value is less than 0.05 there is a significant association between yogurt intake and fluoride exposure in the different cities of Pakistan.

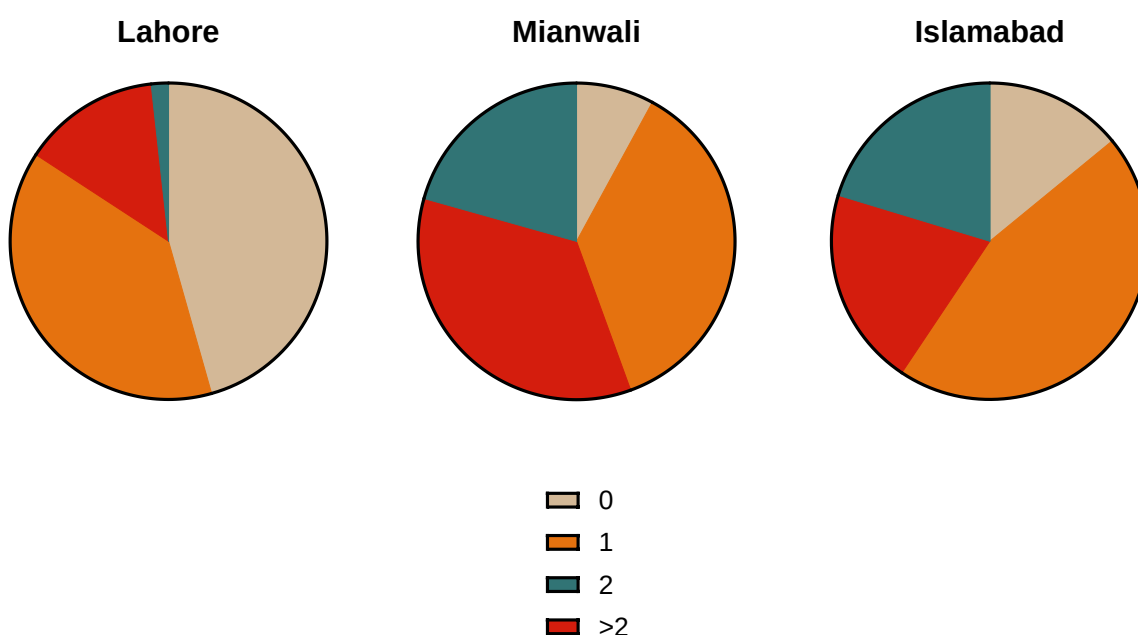
city * No Of Eggs Crosstabulation						
		No Of Eggs				Total
		0	1-2	3-5	>5	
city	Islamabad	11	28	16	9	64
	Lahore	35	14	7	1	57
	Mianwali	31	22	6	4	63
Total		77	64	29	14	184
P Value						<.001



3.4.4. CUPS OF TEA CONSUMED IN A DAY:

Tea intake in Islamabad is highest for people drinking 1 cup (28). Mianwali shows significant representation in the 2 cups and more-than-2 cups categories (22 and 13, respectively). Lahore sees moderate consumption, with 22 people drinking 1 cup. As the P value is less than 0.05 there is a significant association between yogurt intake and fluoride exposure of the different cities of Pakistan.

city * Tea intake Crosstabulation						
		Tea Intake				Total
		0	1	2	>2	
city	Islamabad	11	28	16	9	64
	Lahore	26	22	8	1	57
	Mianwali	5	23	22	13	63
Total		42	73	46	23	184
P Value						<.001

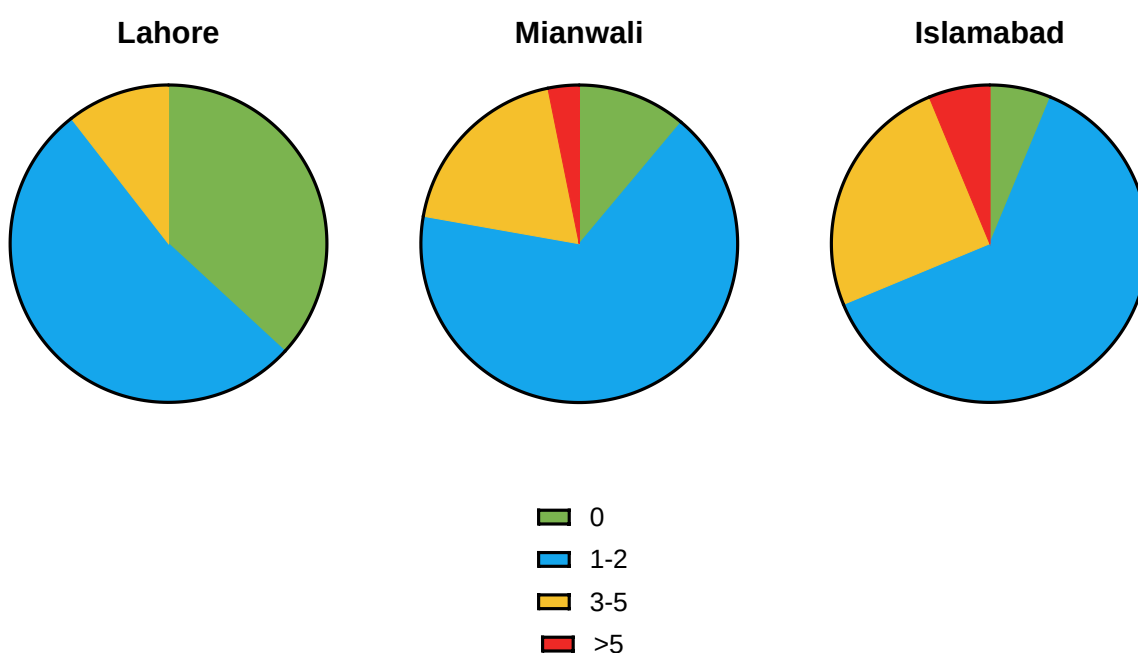


3.4.5. CHICKEN INTAKE PER WEEK:

Chicken consumption is highest in Islamabad, with 40 people consuming 1-2 servings a week. Lahore and Mianwali both show similar consumption levels in the 1-2 servings and 3-5 servings categories. As the P value is less than 0.05 there is a significant association between chicken intake and fluoride exposure of the different cities of Pakistan.

city * chicken Intake Crosstabulation

		Chicken Intake				Total
		0	1-2	3-5	>5	
city	Islamabad	4	40	16	4	64
	Lahore	21	30	6	0	57
	Mianwali	21	34	7	1	63
Total		46	104	29	5	184
P Value						<.001

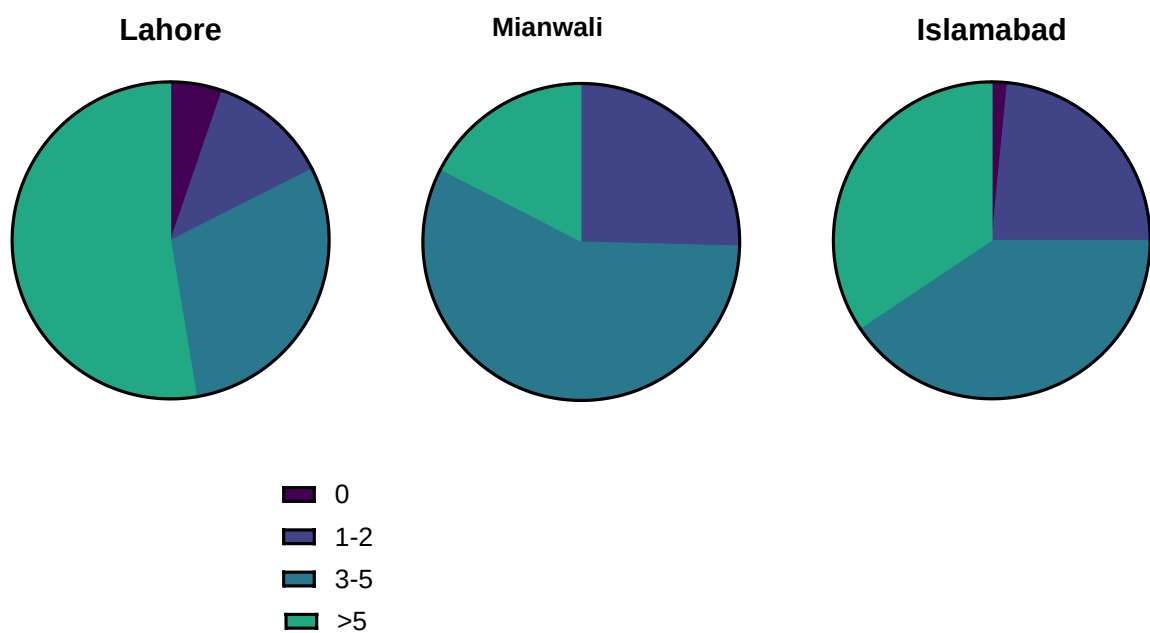


3.4.6. VEGETABLE INTAKE PER WEEK:

Vegetable intake is strong in Islamabad, where most people consume 3-5 servings (26). Lahore stands out for the highest intake of vegetables, with 30 people eating more than 5 servings a week. Mianwali shows a significant number of individuals consuming 3-5 servings (36). As the P value is less than 0.05 there is a significant association between yogurt intake and fluoride exposure of the different cities of Pakistan.

city * vegetable Intake Crosstabulation					
	Vegetable Intake				Total
	0	1-2	3-5	>5	

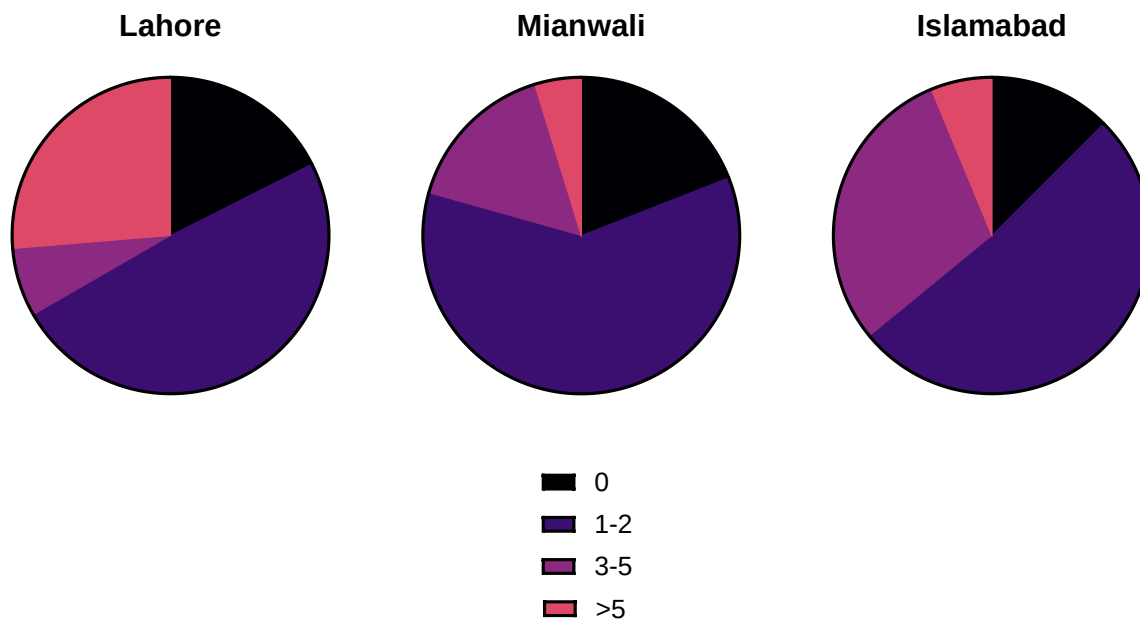
city	Islamabad	1	15	26	22	64
	Lahore	3	7	17	30	57
	Mianwali	0	16	36	11	63
Total		4	38	79	63	184
P Value						<.001



3.4.7. CONSUMPTION OF CARBONATED OR SWEETENED DRINKS:

Sweetened drink consumption is highest in Islamabad, with 33 people consuming 1-2 servings. Lahore and Mianwali show a more balanced distribution across intake levels.

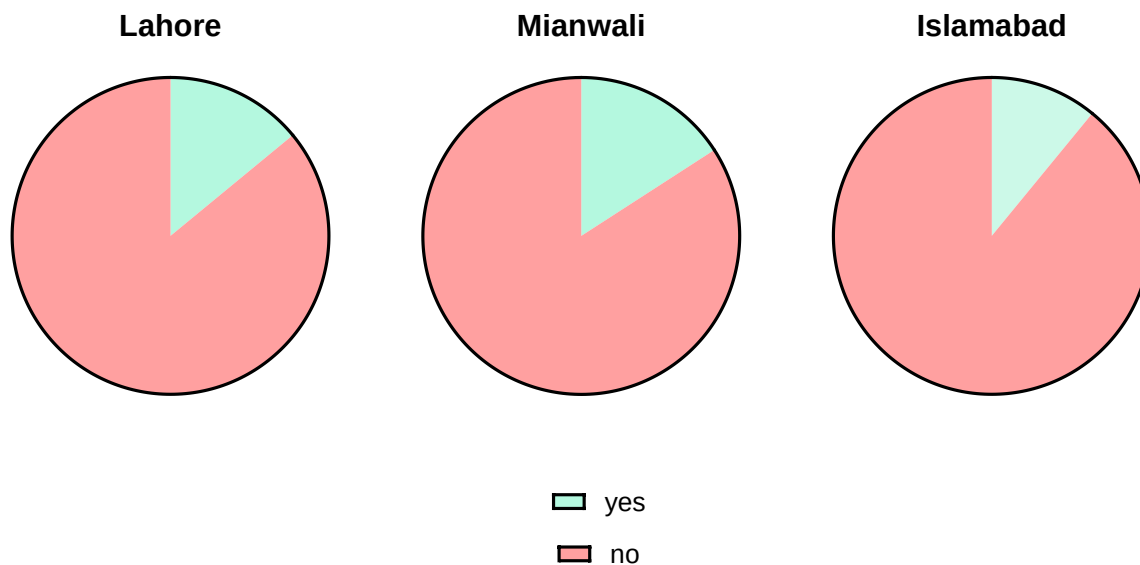
city * Sweetened Drink Intake Crosstabulation						
		Sweetened Drink Intake				Total
		0	1-2	3-5	>5	
city	Islamabad	8	33	19	4	64
	Lahore	9	28	4	16	57
	Mianwali	12	38	10	3	63
Total		29	99	33	23	184



3.4.8. INTAKE OF CALCIUM SUPPLEMENTS:

Calcium supplement use is low across the cities, with Islamabad having 7 users, Lahore having 8, and Mianwali having 10 users. As the P value is more than 0.05 there is no significant association between calcium supplements and fluoride exposure.

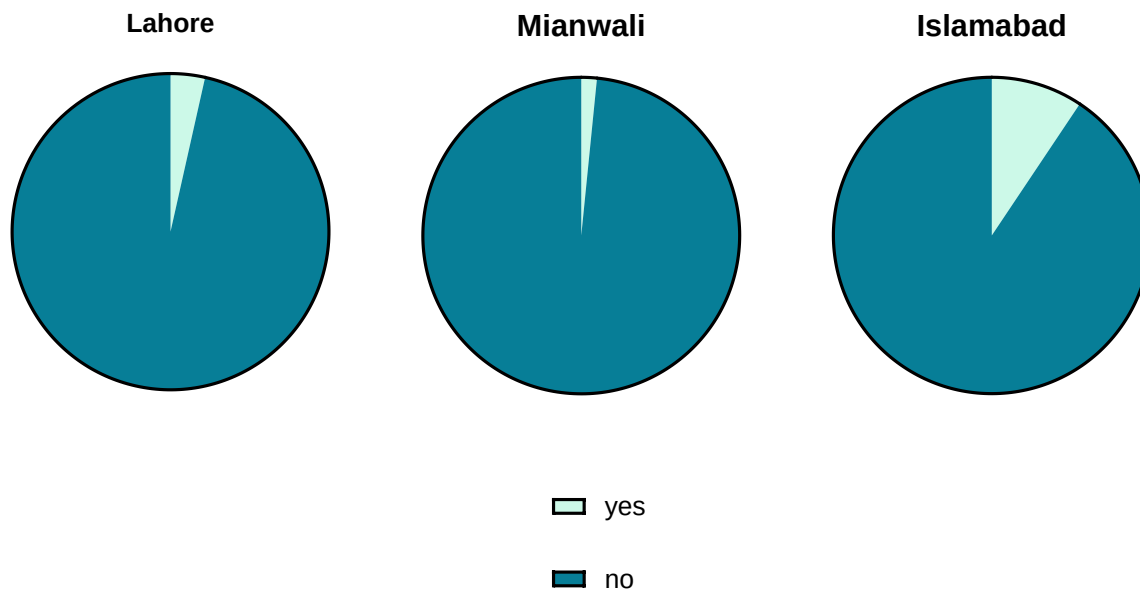
city * Calcium Supplement Crosstabulation				
		Calcium Supplement		Total
		Yes	No	
city	Islamabad	7	57	64
	Lahore	8	49	57
	Mianwali	10	53	63
Total		25	159	184
P Value				0.714



3.4.9. INTAKE OF VITAMIN D SUPPLEMENTS:

Vitamin D supplement intake is also low, with Islamabad having 6 users, and Lahore and Mianwali having just 2 and 1, respectively. As the P value is less than 0.05 there is no significant association between yogurt intake and fluoride exposure in the different cities of Pakistan.

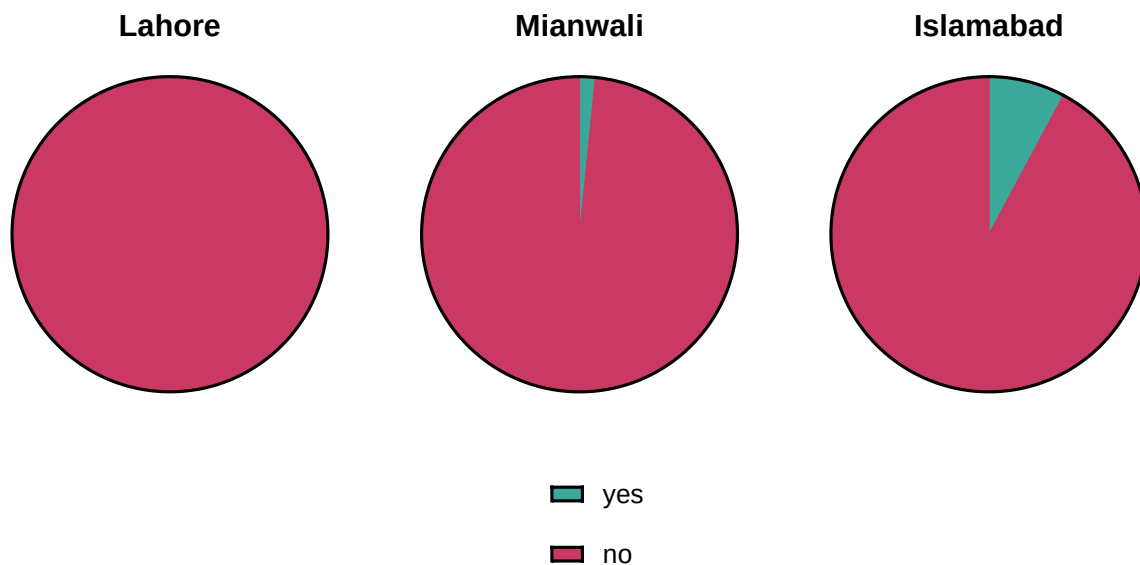
city * Vitamin D Supplement Crosstabulation				
		Vitamin D Supplement		Total
		Yes	No	
city	Islamabad	6	58	64
	Lahore	2	55	57
	Mianwali	1	62	63
Total		9	174	184
P Value				0.153



3.4.9. Multivitamin Supplement Intake by City

Multivitamin supplement use is highest in Islamabad (5 users), while Lahore and Mianwali show very low consumption, with only 0 and 1 user, respectively. The analysis indicates that the P value is greater than 0.05, suggesting that there is no statistically significant association between the use of multivitamins and fluoride exposure. This means that any observed differences in outcomes between individuals who take multivitamins and those who do not are likely due to random chance rather than a meaningful relationship between the two factors. Therefore, we cannot conclude that multivitamin supplementation has any impact on the effects of fluoride exposure based on the current data.

City * Multi-Vitamin Supplement Crosstabulation				
		Multi-Vitamin Supplement		Total
		Yes	No	
city	Islamabad	5	59	64
	Lahore	0	57	57
	Mianwali	1	62	63
Total		6	177	184
P Value				0.065



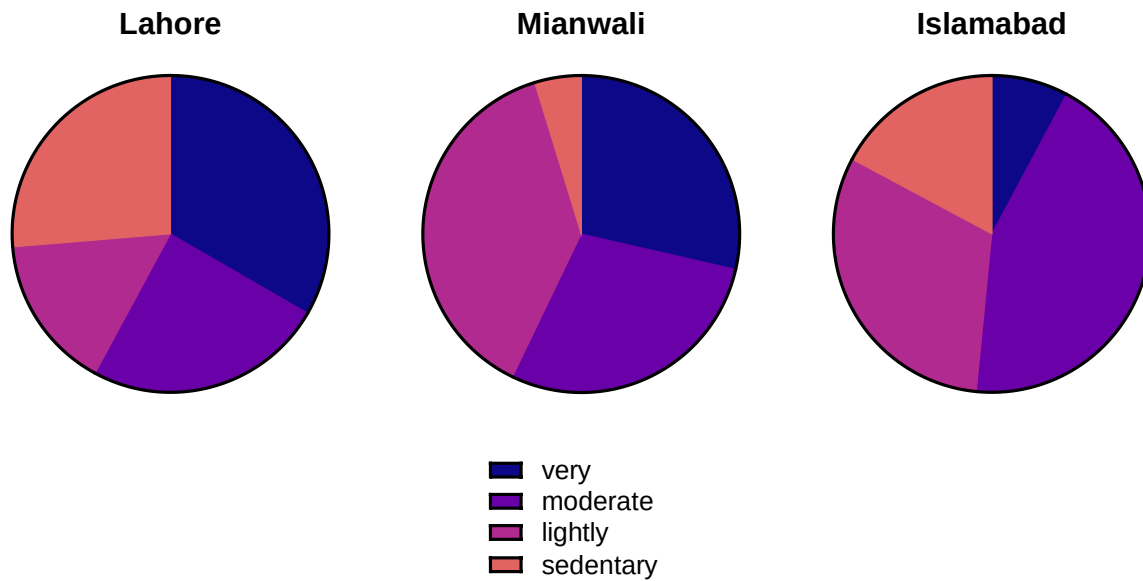
3.5. Health and Lifestyle Patterns Associated with Fluoride Exposure:

3.5.1. Level Of Activity In Normal Day:

Activity levels are moderate in Islamabad, where most people report moderate activity (28). Lahore shows more varied levels of activity, while Mianwali has a significant count of individuals reporting mild activity (24).

As P value is less than 0.05 so there is significant association between level of activity in normal day and fluoride exposure.

city * Level of activity in a normal day Crosstabulation						
		Level of activity in a normal day				Total
		Active	Moderate	Mild	Sedentary	
city	Islamabad	5	28	20	11	64
	Lahore	19	14	9	15	57
	Mianwali	18	18	24	3	63
Total		42	60	53	29	184
		P Value				0.065



3.5.2. Use Of Sunscreen:

Sunscreen use is more common in Islamabad (16 users), with fewer people using sunscreen in Lahore (5) and Mianwali (8).

As P value is less than 0.05 so there is significant association between sunscreen and fluoride exposure.

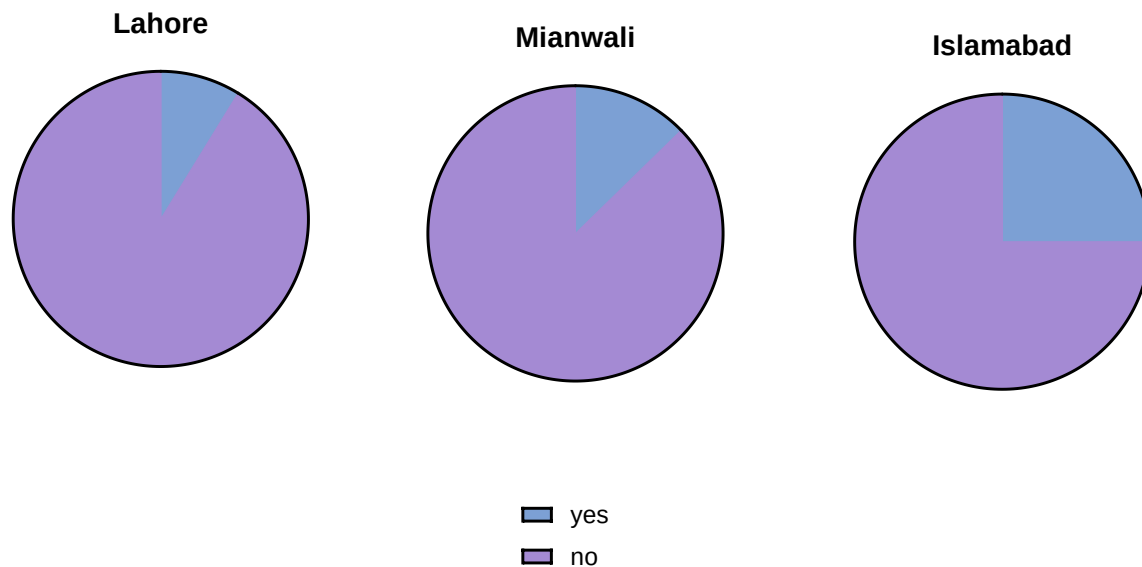


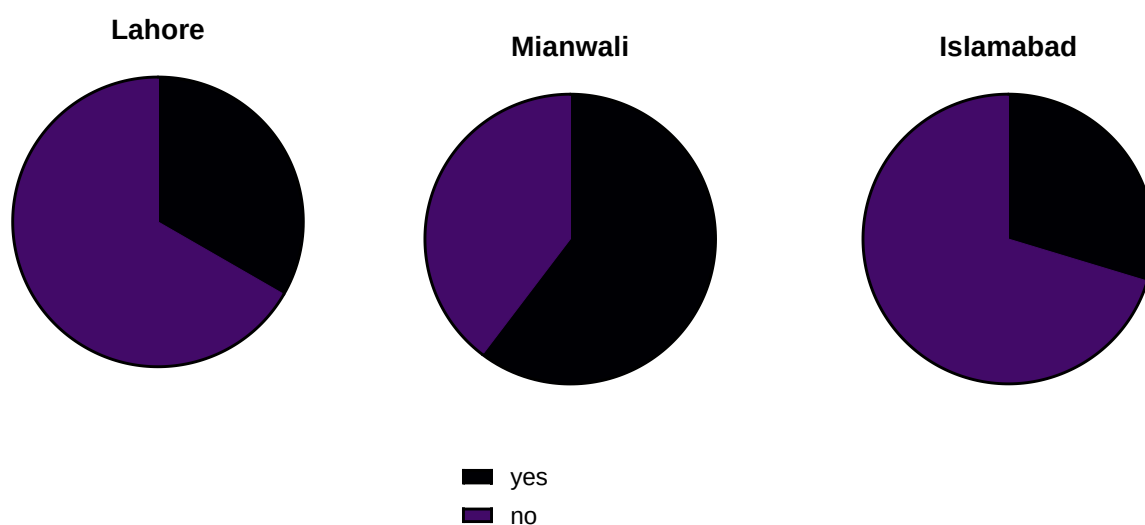
Table:

city * Sunscreen Use Crosstabulation				
		Sunscreen Use		Total
		Yes	No	
city	Islamabad	16	48	64
	Lahore	5	52	57
	Mianwali	8	55	63
Total		29	155	184
P Value				0.065

3.5.3. Activity In Sunlight for Longer Than Hour:

Mianwali has the highest number of people reporting activity in sunlight (38), while Islamabad and Lahore each report 19 individuals. As P value is less than 0.05 so there is significant association between activity in sunlight and fluoride exposure. As P value is less than 0.05 so there is significant association between activity in sunlight and fluoride exposure.

city * Activity in Sunlight Crosstabulation				
		Activity in Sunlight		Total
		Yes	No	
City	Islamabad	19	45	64
	Lahore	19	38	57
	Mianwali	38	25	63
Total		76	108	184
P Value				<.001

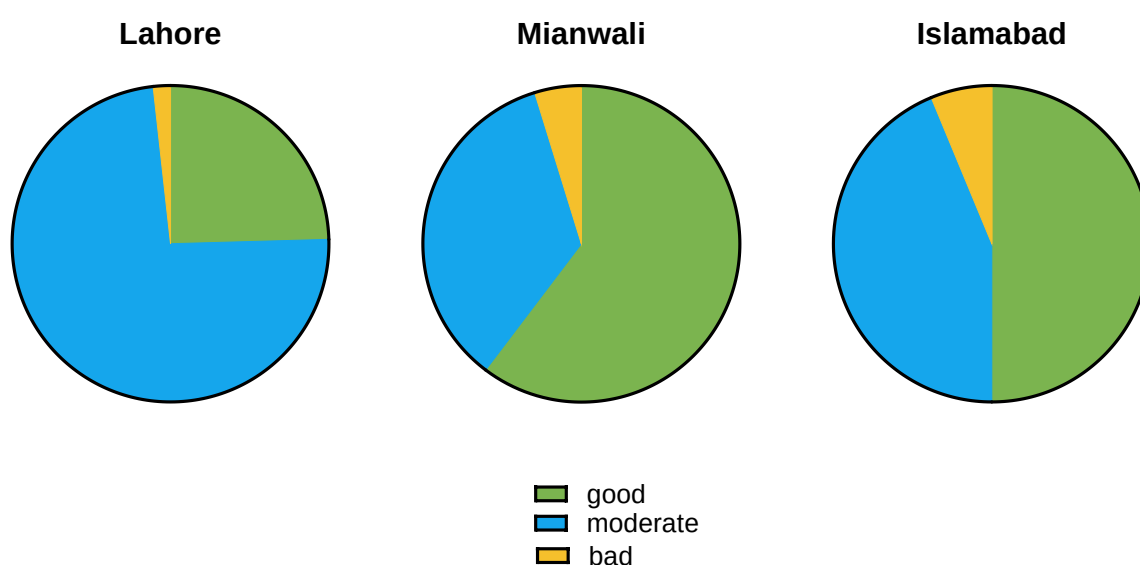


3.5.4. Dental Health Subjects Pov:

Islamabad has the highest count of individuals reporting "Good" dental health (32), while Lahore shows a higher number of people with "Mediocre" dental health (28). Mianwali is more evenly spread across all categories but has a notable count of people with "Good" dental health (39). As P value is less than 0.05 so there is significant association between dental health and fluoride exposure.

city * Dental Health Crosstabulation						
	Dental Health					Total
	Good	Moderate	Mediocre	Mild	Bad	

city	Islamabad	32	11	17	4	0	64
	Lahore	13	15	28	0	1	57
	Mianwali	39	4	6	11	3	63
Total		84	30	50	15	4	184
P Value							<0.001

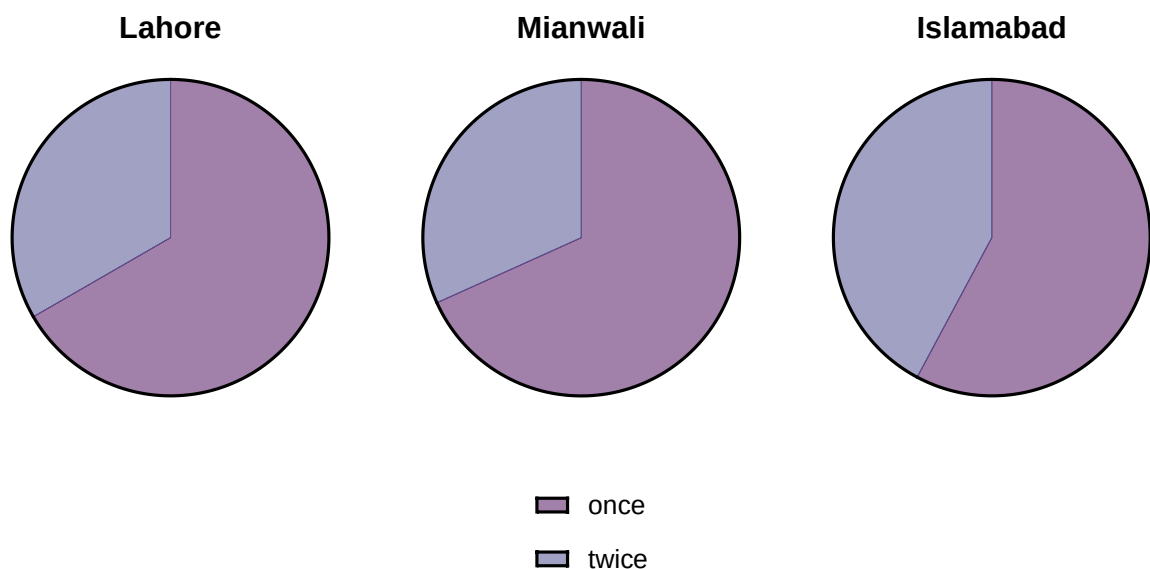


3.5.5. Number Of Times Brushing Teeth Daily:

Islamabad stands out for having more people brushing twice daily and showing "Normal" fluorosis (25). In Lahore, those with "Moderate" fluorosis are more likely to brush once a day (17), while Mianwali shows similar brushing habits for individuals in the "Questionable" category.

Level of Fluorosis * No. of times Brushing Crosstabulation				
	No. of times Brushing			Total
	Islamabad	Lahore	Mianwali	

		1	2	1	2	1	2	
Level of Fluorosis	Normal	36	28	8	10	20	11	113
	Moderate	1	1	17	4	2	2	27
	Mild	1	2	3	0	3	1	13
	Very Mild	4	1	6	3	7	4	25
	Severe	1	0	4	2	2	0	9
	Questionable	3	2	5	3	8	3	24
Total		46	34	43	22	42	21	184

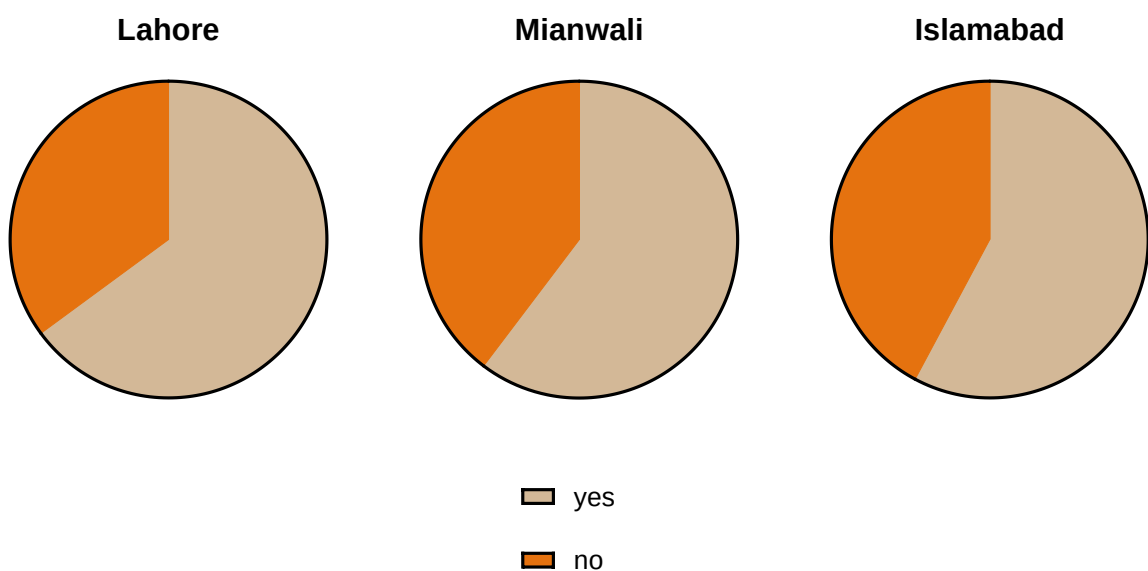


3.5.6. Experienced Toothache:

Islamabad records the highest number of pain cases in the "Normal" category (32). In contrast, Lahore's pain cases are more common among those with "Moderate" fluorosis (4). Mianwali shows notable pain cases in the "Severe" (6) and "Questionable" categories.

Level of Fluorosis * Pain in Teeth Crosstabulation		
	Pain in Teeth	Total

		Islamabad		Lahore		Mianwali		
		Yes	No	Yes	No	Yes	No	
Level of Fluorosis	Normal	36	28	2	2	4	1	73
	Moderate	2	1	4	3	18	7	35
	Mild	2	1	15	6	2	2	28
	Very Mild	3	1	2	1	0	4	11
	Severe	1	0	6	3	6	5	21
	Questionable	3	2	4	2	0	2	13
Total		47	33	33	17	30	21	184

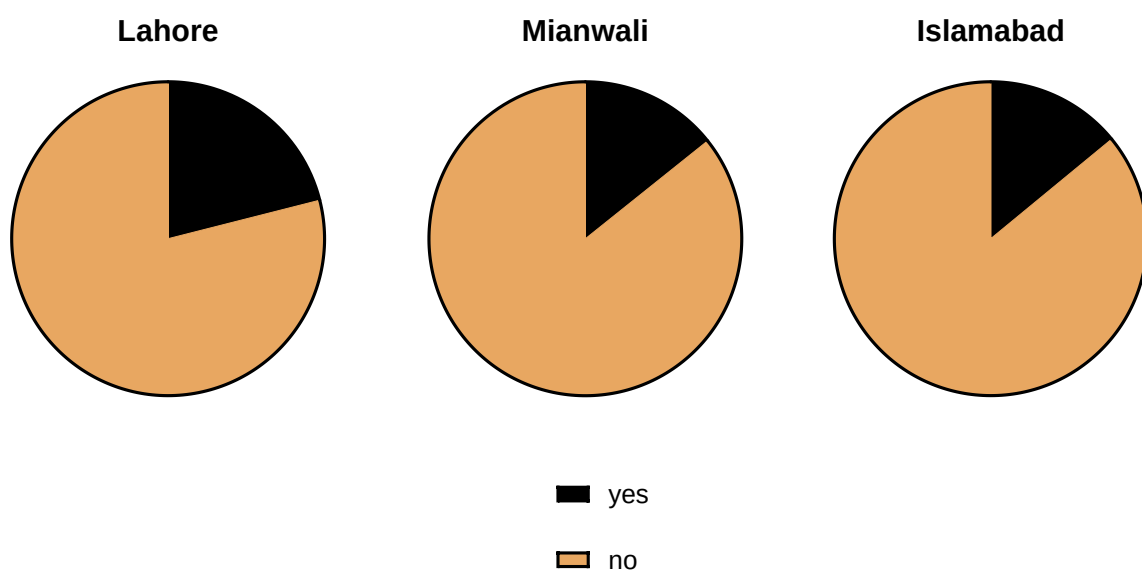


3.5.7. VISIT DENTIST FOR REGULAR CHECKUPS:

Dentist visits are less frequent in Islamabad, with only 8 visits in the "Normal" category. Lahore records fewer dentist visits in the "Moderate" category (4), and Mianwali sees a higher count of individuals not visiting the dentist, especially in the "Questionable" category.

Level of Fluorosis * Visit Dentist Crosstabulation										
				Visit Dentist				Total		
				Islamabad		Lahore			Mianwali	
				Yes	No	Yes	No		Yes	No

Level of Fluorosis	Normal	8	40	2	5	2	23	99
	Moderate	0	0	4	17	0	4	25
	Mild	0	1	0	3	0	4	8
	Very Mild	0	2	0	9	4	7	22
	Severe	0	0	4	2	0	2	8
	Questionable	1	2	1	6	3	9	22
Total		9	64	11	42	9	49	184



CHAPTER 4
DISCUSSION

1.4. Overview of Prevalence Rates:

A study on the prevalence rate among adolescents aged 12 to 16 years in Mianwali and Lahore with a control group was conducted in Islamabad. These findings present wide discrepancies amongst the regions. Some varied results were found in districts: 90% with varying pathologies of dental fluorosis was found in Mianwali, ranging from very mild to moderate, as well as some cases showing evidence of severe fluorosis; Lahore showed a prevalence of 43%, while its breakdown into very mild of 25%, mild, at 15%, and another moderate 5%, with a nil occurrence of severe types of fluorosis. Accordingly, Islamabad showed much-lowered prevalence rates, consistent even to averages compared internationally, thus indicating its own relatively lower fluoride exposure accordingly.

1.5. Dental Fluorosis: Implications

Dental fluorosis is a disease that affects the enamel of the teeth by result of long term exposure to high concentration of fluoride during the tooth formation stage. In adolescent children, it is not just about physical appearance, but one has to consider issues to do with feelings of self-worth and health of the heart. Mild forms of fluorosis are often symptomless, but as the severity increases, this can lead to social embarrassment and distress, with consequent impairment of quality of life.

1.6. Key Risk Factors Identified:

The following are different risk factors that have brought regional variations in the prevalence of dental fluorosis:

- **High Fluoride Levels in Drinking Water:** Mianwali has very high levels of natural fluoride in groundwater, way above the recommended permissible limits, and is thus a major risk factor.
- **Improper Use of Fluoride Products:** In Lahore, improper use of fluoride-containing toothpaste and mouthwashes without proper guidance became a major contributing factor for the prevalence.

- **Sources of Fluoride in Diet:** Tea, foodstuffs, and other edible items prepared with the addition of fluoride-rich water augmented the intake of fluoride in both regions.
- **Socio-economic Disparities:** In Mianwali, poor access to fluoride-free drinking water and dental education was an aggravating factor. In Lahore, the higher intake of fluoride was because of the urban way of life, not because of lack of resources.

Comparison with Islamabad thus provides an important example of how proper management of fluoride exposure is carried out. A better regulation of drinking water, a higher level of awareness, and better socioeconomic conditions led to the lowest rates of fluorosis in Islamabad; it therefore provides an effective control group for the study.

4.4. Recommendations for Public Health Interventions:

Based on the severity of dental fluorosis throughout, the following suggestions have come forth for the public health interventions:

- **Mianwali:** Reduction in fluoride level of the ground waters by bringing better filtering mechanisms and additional sources of water.
- **Lahore:** Propaganda and educational drives broadcasting the proper usage of fluoride, adapting to the correct brushing method, and awareness about the contents in various dental products.
- **General:** The formulation of national policies regarding the regulation of fluoride concentrations in water supplies and education of the general public about fluoride exposure.

4.5. Balancing Benefits and Risks of Fluoride:

This study highlights the need for a balance between the preventive benefits of fluoride and the risks of overexposure. Policymakers and health professionals must collaborate to ensure that fluoride exposure is optimized to prevent dental caries with minimal risk of fluorosis. Since the interventions deal with the very causes of overexposure to fluoride, improvement in oral health, psychological well-being, and the overall quality of life of adolescents in areas contaminated by fluoride may thus be achieved.

CHAPTER 5

CONCLUSION

Prevalence of dental fluorosis was this study studied on adolescents (12 to 16 years) residing in Mianwali and Lahore, and juxtaposed against a control group of Islamabad. The findings showed that there was also a high prevalence in Mianwali, in part explained by raised fluoride levels in groundwater, as well as moderate prevalence in Lahore attributed largely to inappropriate use of fluoride containing dental products. The rates remarkably showed much lower in Islamabad, this depicts the impact of balanced fluoride exposure and better public awareness.

These results highlight the need for region specific public health interventions. Mianwali needs improved water filtration systems to lower its fluoride, and Lahore could learn from educational campaigns on how to properly use its fluoride products. Guidelines are necessary to be developed by policymakers regarding the optimal amount of fluoride exposure – to prevent dental caries with a minimum risk of fluorosis. We can improve oral health and the overall well-being of adolescents in affected areas by addressing these challenges.

CHAPTER 6

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